Chapter 10. Generative Models for Classification

Lecture 7-3 (Bayes theorem, LDA, QDA, naive Bayes, logistic regression vs LDA, comparison of classifiers)

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1 Discriminant Analysis as Generative Modeling

Generative models describe

$$\Pr(X, Y) = \Pr(X \mid Y)\Pr(Y)$$

and classify by Bayes' rule. Discriminant analysis:

(1) model X in each class, (2) flip with Bayes' theorem. Gaussian classes:

$$X \mid Y = c_k \sim N(\mu_k, \Sigma) \;\; \text{(LDA)}, \qquad X \mid Y = c_k \sim N(\mu_k, \Sigma_k) \;\; \text{(QDA)};$$

1.1 Bayes' theorem for classification

With class densities f_k and priors $\pi_k = \Pr(Y = k)$,

$$p_k(x) = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)} \;\propto\; \pi_k f_k(x).$$

$\hat{C}(x) = \arg\max_k \pi_k f_k(x)$ (the Bayes classifier of Ch. 8, estimated). Favoring a class moves the boundary away from it.

1.2 Why discriminant analysis?

Logistic regression is unstable for well-separated classes (Exercise 9.1); LDA is not, and it is also stabler for small n with near-normal classes. It suits $K > 2$ and gives low-dimensional views.

2 Linear Discriminant Analysis

Motivation. Model each class density as normal and apply Bayes' rule.

2.1 One predictor

$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\{-\frac{1}{2}(\frac{x-\mu_k}{\sigma_k})^2\}$ with $\sigma_1 = \cdots = \sigma_K = \sigma$:

$$p_k(x) = \frac{\pi_k \exp\{-\frac{1}{2\sigma^2}(x-\mu_k)^2\}}{\sum_{l=1}^{K} \pi_l \exp\{-\frac{1}{2\sigma^2}(x-\mu_l)^2\}}.$$

Theorem: discriminant functions

Classifying to the largest $p_k(x)$ is equivalent to classifying to the largest

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k,$$

a *linear* function of x. With $K = 2$ and $\pi_1 = \pi_2 = 0.5$ the decision boundary is $x = (\mu_1 + \mu_2)/2$.

Proof [Supplement]. The denominator does not depend on k. Taking logs, $\log(\pi_k f_k(x)) = \log \pi_k - \frac{(x-\mu_k)^2}{2\sigma^2} + c = \log \pi_k - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + c$, and $-x^2/(2\sigma^2) + c$ is common to all classes. For two classes with equal priors, $\delta_1(x) = \delta_2(x)$ gives $x(\mu_1 - \mu_2) = (\mu_1^2 - \mu_2^2)/2$. ■

Example: simulation

$\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, $\sigma^2 = 1$: Bayes boundary $x = 0$, Bayes error $\Phi(-1.5) \approx 6.7\%$ (Exercise 8.2). With 20 observations per class the LDA boundary is close to 0 and its error slightly above.

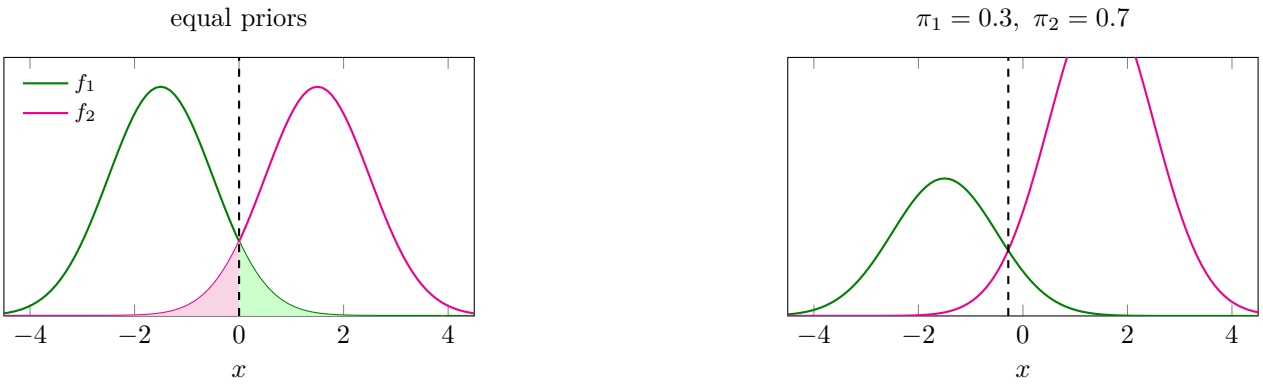


Figure 1: $\pi_k f_k$ for $\mu_{1,2} = \mp 1.5$, $\sigma = 1$. Right: the boundary moves to $\frac{\mu_1+\mu_2}{2} + \frac{\sigma^2 \log(\pi_1/\pi_2)}{\mu_2-\mu_1} \approx -0.28$.

2.2 Estimating the parameters

$$\hat{\pi}_k = \frac{n_k}{n}, \qquad \hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i, \qquad \hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2 \quad \text{(pooled variance)},$$

the full-likelihood MLEs except for the divisor $n - K$ (Exercise 10.2).

2.3 Several predictors

$$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\Big\{ -\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big\},$$

with $\Sigma_k = \Sigma$ for all k.

Theorem: LDA in p dimensions

$$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k, \qquad \hat{C}(x) = \arg\max_k \delta_k(x).$$

The boundary between classes k and k' is the hyperplane

$$\Big\{x:\ x^\top \Sigma^{-1}(\mu_k - \mu_{k'}) + \log\frac{\pi_k}{\pi_{k'}} - \frac{1}{2}(\mu_k + \mu_{k'})^\top \Sigma^{-1}(\mu_k - \mu_{k'}) = 0\Big\},$$

with normal vector $\Sigma^{-1}(\mu_k - \mu_{k'})$, generally *not* in the direction of $\mu_k - \mu_{k'}$.

Proof [Supplement]. As in one dimension: $\log(\pi_k f_k(x)) = \log \pi_k - \frac{1}{2}(x-\mu_k)^\top \Sigma^{-1}(x-\mu_k) + c$, and the quadratic term $-\frac{1}{2}x^\top \Sigma^{-1} x$ is shared by all classes because Σ is common. Set $\delta_k = \delta_{k'}$. ■

Iris data (4 variables, 3 species, 50 each): LDA misclassifies 3 of 150 training samples. **Default data**: training error 2.75%, FPR 0.2%, FNR 75.7% (Ch. 8).

2.4 Adjusting the errors

LDA uses threshold 0.5 on $\Pr(\text{Yes} \mid \text{balance}, \text{student})$. Lowering it to 0.2 cuts the Default FNR from 75.7% to about 41% with a slightly higher overall error; likewise for logistic regression.

3 Quadratic Discriminant Analysis

Class-specific Σ_k:

$$\delta_k(x) = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^\top \Sigma_k^{-1}(x-\mu_k) + \log\pi_k = x^\top W_k x + x^\top w_k + b_k,$$

with $W_k = -\frac{1}{2}\Sigma_k^{-1}$, $w_k = \Sigma_k^{-1}\mu_k$ and $b_k = -\frac{1}{2}\mu_k^\top\Sigma_k^{-1}\mu_k - \frac{1}{2}\log|\Sigma_k| + \log\pi_k$: a quadratic form. The boundary between k and k' is the quadric

$$\{x:\ x^\top(W_k - W_{k'})x + x^\top(w_k - w_{k'}) + (b_k - b_{k'}) = 0\}.$$

Parameter count. Covariance parameters: LDA $p(p+1)/2$, QDA $Kp(p+1)/2$ (1275 versus 3825 for $p = 50$, $K = 3$). QDA suits clearly different covariances and enough data; LDA suits similar covariances or few data (lower variance, possibly higher bias).

4 Naive Bayes

Inputs independent *within each class*:

$$f(x_1, \ldots, x_p \mid y = c_k) = f(x_1 \mid c_k) f(x_2 \mid c_k) \cdots f(x_p \mid c_k).$$

Gaussian naive Bayes takes each Σ_k diagonal:

$$\delta_k(x) \ \propto\ \log\Big[\pi_k \prod_{j=1}^{p} f_{kj}(x_j)\Big] = -\frac{1}{2}\sum_{j=1}^{p}\Big[\frac{(x_j-\mu_{kj})^2}{\sigma_{kj}^2} + \log\sigma_{kj}^2\Big] + \log\pi_k.$$

Only one-dimensional densities are needed (Gaussian, kernel, or frequency tables). It often works when n is not large relative to p: the bias is outweighed by the variance reduction, and only the *ranking* of posteriors matters.

5 Comparison of Classification Methods

5.1 Logistic regression versus LDA

LDA log posterior odds:

$$\log\frac{\Pr(y=c_k \mid x)}{\Pr(y=c_K \mid x)} = x^\top\Sigma^{-1}(\mu_k - \mu_K) + \log\frac{\pi_k}{\pi_K} - \frac{1}{2}(\mu_k+\mu_K)^\top\Sigma^{-1}(\mu_k-\mu_K) = \alpha_{k0} + \alpha_k^\top x,$$

and logistic regression has linear logits $\beta_{k0} + \beta_k^\top x$ by construction. They are different estimators:

- **Origin of linearity.** For LDA a consequence of Gaussian classes with common covariance.
- **Common component.** In $\Pr(X=x, Y=c_k) = \Pr(X=x)\Pr(Y=c_k \mid x)$ both have the multinomial-logit second factor $\exp(\beta_{k0}+\beta_k^\top x)/(1+\sum_{k'}\exp(\beta_{k'0}+\beta_{k'}^\top x))$.
- **Different assumptions on $\Pr(X)$.** Logistic: arbitrary; LDA: the mixture $\sum_k \pi_k \phi(x;\mu_k,\Sigma)$.

- **Estimation.** Logistic maximizes the *conditional* likelihood (marginal left nonparametric, mass $1/n$ per point); LDA the *full* likelihood of $\phi(x; \mu_k, \Sigma)\pi_k$.

Remarks. Under Gaussian classes logistic regression may lose about 30% asymptotic efficiency in error rate (Efron, 1975). LDA uses all points to estimate Σ (outlier sensitive); logistic regression down-weights points far from the boundary (tiny $p_i(1-p_i)$).

5.2 QDA versus LDA, logistic regression and KNN

Linear true boundary: LDA and logistic regression; moderately nonlinear: QDA; complicated: KNN with K by CV, given enough data.

Method	Boundary	Strengths / weaknesses
logistic regression	linear (or by basis expansion)	few assumptions; probabilities; unstable under separation
LDA	linear	efficient under Gaussian classes; stable with small n; outlier sensitive
QDA	quadratic	flexible covariances; many parameters
naive Bayes	depends on f_{kj}	scales to large p; independence assumption
KNN	arbitrary	nonparametric; needs much data; no interpretation

Summary

- Classify by $\arg\max_k \pi_k f_k(x)$: LDA linear, QDA quadratic, naive Bayes trades bias for variance.
- LDA and logistic regression share the logit form but differ in $\Pr(X)$ and in full versus conditional likelihood.

6 Selected exercises

Motivation. ISLR Ex. 4.3 and 4.7, and the MLE property of the LDA estimates.

Exercise 10.1 (QDA in one dimension; ISLR Ex. 4.3)

Let $X \mid Y = k \sim N(\mu_k, \sigma_k^2)$ with priors π_k, $k = 1, \ldots, K$.
(a) Show that the Bayes classifier assigns x to the largest

$$\delta_k(x) = -\frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} - \log\sigma_k + \log\pi_k.$$

(b) Show that if $\sigma_k \neq \sigma_l$, then $\delta_k - \delta_l$ is a genuine quadratic, the two classes are separated by at most two points, and the class with the larger variance wins in both tails; so the Bayes classifier is not linear.

Solution.

(a) The Bayes classifier maximizes $\pi_k f_k(x)$, equivalently its logarithm

$$\log\pi_k - \tfrac{1}{2}\log(2\pi) - \log\sigma_k - \frac{(x-\mu_k)^2}{2\sigma_k^2}.$$

Drop the common $-\frac{1}{2}\log(2\pi)$ and expand; unlike LDA, $-x^2/(2\sigma_k^2)$ depends on k.

(b) $\delta_k(x) - \delta_l(x) = Ax^2 + Bx + C$ with $A = \frac{1}{2\sigma_l^2} - \frac{1}{2\sigma_k^2} \neq 0$. A nonzero quadratic has at most two real roots, so the line splits into at most three intervals, alternately assigned to k and l. If $\sigma_k > \sigma_l$, then $A > 0$ and $\delta_k - \delta_l \to +\infty$ as $|x| \to \infty$: class k wins on both tails, which no single threshold (linear rule) can produce. ■

Exercise 10.2 (LDA estimates are maximum-likelihood estimates)

Observe (x_i, y_i), $i = 1, \dots, n$, with $x_i \in \mathbb{R}$, $y_i \in \{1, \dots, K\}$, class sizes $n_k \geq 1$, and full likelihood $\prod_i \pi_{y_i}\phi(x_i; \mu_{y_i}, \sigma^2)$. Assume $W = \sum_k \sum_{i:y_i=k}(x_i - \bar{x}_k)^2 > 0$, $\bar{x}_k$ the class means.
(a) Show that the MLE is $\hat{\pi}_k = n_k/n$, $\hat{\mu}_k = \bar{x}_k$, $\hat{\sigma}^2_{\text{ML}} = W/n$.
(b) Show that $\mathbb{E}W = (n-K)\sigma^2$, so the pooled $\hat{\sigma}^2 = W/(n-K)$ is the unbiased rescaling.

Solution [Supplement].

(a) The log-likelihood separates:

$$\sum_k n_k \log \pi_k \;-\; \frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_k \sum_{i:y_i=k}(x_i - \mu_k)^2.$$

Step 1 (π). For $\pi_k > 0$, $\sum_k \pi_k = 1$, Jensen's inequality for the concave log gives

$$\sum_k n_k \log \pi_k - \sum_k n_k \log \frac{n_k}{n} = n \sum_k \frac{n_k}{n} \log \frac{n\pi_k}{n_k} \leq n \log \sum_k \pi_k = 0,$$

with equality iff $n\pi_k/n_k$ is constant, i.e. $\pi_k = n_k/n$.

Step 2 (μ). For any σ^2 the last term is maximized by minimizing each $\sum_{i:y_i=k}(x_i - \mu_k)^2 = \sum_{i:y_i=k}(x_i - \bar{x}_k)^2 + n_k(\bar{x}_k - \mu_k)^2$, i.e. at $\mu_k = \bar{x}_k$; the minimum value of the double sum is W.

Step 3 (σ^2). $s \mapsto -\frac{n}{2}\log s - \frac{W}{2s}$ has derivative $(W - ns)/(2s^2)$, positive for $s < W/n$ and negative beyond, so the maximum is at W/n.

(b) Within class k the x_i are i.i.d. $N(\mu_k, \sigma^2)$, so $\mathbb{E}\sum_{i:y_i=k}(x_i - \bar{x}_k)^2 = (n_k - 1)\sigma^2$; summing over k gives $(n-K)\sigma^2$. ■

Exercise 10.3 (Predicting a dividend; ISLR Ex. 4.7)

Among companies that issued a dividend (80% of them) last year's percentage profit X has mean 10; among the others, mean 0; both have variance 36, and X is normal within each group. Find the probability that a company with $X = 4$ issues a dividend.

Solution. By Bayes' theorem with $f_k(x) \propto \exp\{-(x - \mu_k)^2/72\}$ (the common factor $1/\sqrt{72\pi}$ cancels),

$$\Pr(\text{Yes} \mid X = 4) = \frac{0.8\, e^{-(4-10)^2/72}}{0.8\, e^{-(4-10)^2/72} + 0.2\, e^{-(4-0)^2/72}} = \frac{0.8\, e^{-1/2}}{0.8\, e^{-1/2} + 0.2\, e^{-2/9}} \approx \frac{0.4852}{0.6454} \approx 0.752.$$

$X = 4$ is closer to 0, but the prior 0.8 tips the posterior. ■